



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
UNIT TEST
28-06-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

Time : 3 Hours

Maximum Marks : 216

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

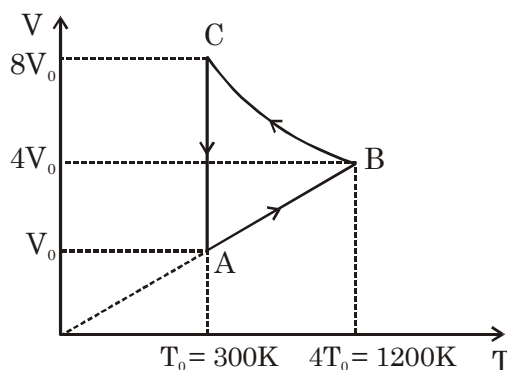
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

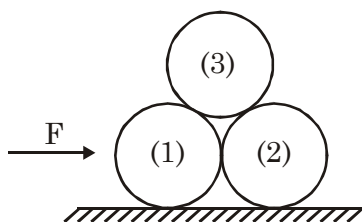
- Two moles of an ideal mono-atomic gas is taken through a cyclic process ABCA as shown in the V-T diagram. In the process BC, $TV^2 = \text{constant}$. (Take : $\ln(2) = 0.7$)



- Work done by the gas is maximum in process AB.
- Work done by the gas in process BC is $900R$.
- Heat supplied to the system in process AB is $4500R$.
- Efficiency of the cycle is greater than 25%.

Space for Rough Work

2. Three identical cylinders are arranged in a triangle as shown in figure, with the bottom two lying on the ground. The ground and the cylinders are frictionless. You apply a constant horizontal force (directed to the right) on the left cylinder.

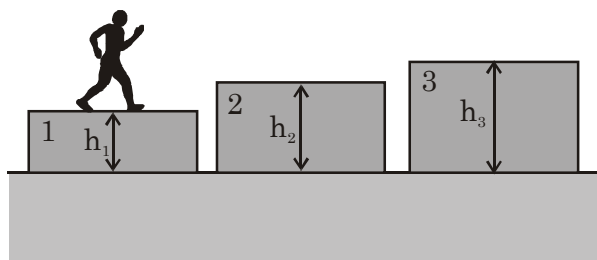


N_{12} is normal reaction between (1) and (2), N_{13} is normal reaction between (1) and (3), N_{23} is normal reaction between (2) and (3)

- (A) If $F = \frac{mg}{2\sqrt{3}}$, then N_{12} will become zero.
- (B) If $F = 2\sqrt{3} mg$, then N_{23} will become zero.
- (C) If $F = \sqrt{3} mg$ then N_{13} will become zero.
- (D) If $F = \frac{mg}{\sqrt{3}}$, all cylinder will have same acceleration but N_{12} will be zero.

Space for Rough Work

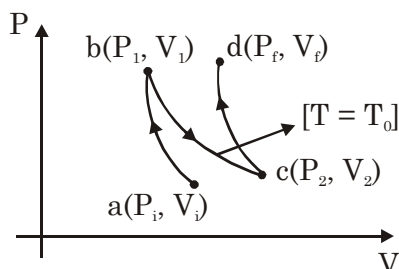
3. Three icebergs of shape of cuboid of same surface area at the top are floating in the salty ocean. When a man is standing on iceberg 1, the free surface height from water surface are equal to $h_1 = 5$ cm, $h_2 = 10$ cm and $h_3 = 12$ cm respectively. When he switched from iceberg 1 to iceberg 2 then in equilibrium the surface of icebergs 1 and 2 were on same height above water level. After that, the man switched from iceberg 2 to iceberg 3. When the iceberg are in now in equilibrium, what are the corresponding heights?



- (A) $h_1' = 7.5$ cm
 (B) $h_2' = 12$ cm
 (C) $h_3' = 9.5$ cm
 (D) the common heights of icebergs 1 and 2 above water level when man was standing on 2 was 7.5 cm
4. One mole of an ideal gas undergoes through the process $T = T_0 + kV^3$ where T_0 and k are positive constants and V is volume. Then which of the following is/are correct.
- (A) The volume for which pressure will be minimum is $\left(\frac{T_0}{2k}\right)^{1/3}$.
 (B) The volume for which pressure will be minimum is $\left(\frac{T_0}{2k}\right)^{2/3}$.
 (C) The minimum pressure attainable is $\frac{3}{2}(2^{1/3})Rk^{1/3}T_0^{2/3}$.
 (D) The minimum pressure attainable is $\frac{3}{2^{2/3}}Rk^{1/3}T_0^{2/3}$.

Space for Rough Work

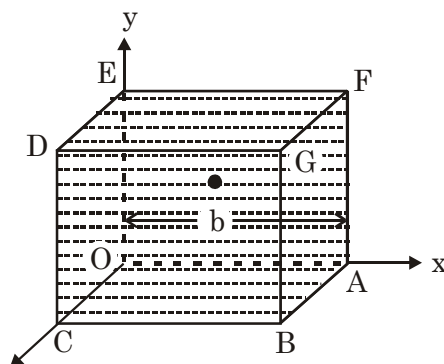
5. One mole of an ideal monochromatic gas in an initial state a with pressure P_i and volume V_i is to be taken to a final state d with $P_f = B^2 P_i$ and $V_f = V_i/B$ through the path $a \rightarrow b \rightarrow c \rightarrow d$ as shown in figure below for a particular values of $B(> 1)$. Here $a \rightarrow b$ and $c \rightarrow d$ are adiabatic paths while $b \rightarrow c$ is an isothermal with temperature T_0 . States b and c correspond to (P_1, V_1) and (P_2, V_2) respectively.



- (A) The ratio of $\frac{V_2}{V_1}$ is $\sqrt{B}$.
- (B) The ratio of $\frac{V_2}{V_1}$ is $\sqrt{B^3}$.
- (C) Work done by the gas $\frac{RT_0}{2} \ln B + \frac{3}{2} P_i V_i (1 - B)$.
- (D) Work done by the gas $\frac{RT_0}{2} \ln B + \frac{3}{2} (RT_0 - B P_i V_i)$.

Space for Rough Work

6. A liquid of density ρ is filled in a cubical container OABCDEFG. The point O is assume to be origin. A small ball of density ρ_0 is at $\left(\frac{b}{2}, \frac{b}{2}, \frac{b}{2}\right)$ as shown in figure and acceleration $2g\hat{i} + g\hat{j}$ is given to container. Uniform gravity exists along negative y-axis. Now, choose the correct option(s)



- (A) If $\rho = \rho_0$ then acceleration of small ball is $\sqrt{5}g$.
 (B) If $\rho = 2\rho_0$ then small ball will have acceleration of $5g$.
 (C) If $\rho = 3\rho_0$ then small ball will hit the mid of edge GF.
 (D) If $\rho = \rho_0$ then net force on the small ball is zero.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 16)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 7 and 8

Imagine a system, that can keep the room temperature within a narrow range between 20°C to 25°C . The system includes a heat engine operating with variable power $P = 3 KT$, where K is a constant coefficient, depending upon the thermal insulation of the room, the area of the walls and the thickness of the walls. T is temperature of the room in degree. When the room temperature drops lower than 20°C , the engine turns on, when the temperature increase over 25°C , the engine turns off, room loses energy at a rate of $K(T - T_0)$. T_0 is the outdoor

temperature. The heat capacity of the room is C . (Given : $T_0 = 10^{\circ}\text{C}$, $\ln\left(\frac{3}{2}\right) = 0.4$, $\ln\left(\frac{6}{5}\right) = 0.18$,

$$\frac{C}{K} = 750 \text{ SI unit}).$$

7. Suppose at $t = 0$, the engine turns off, after how much time interval, again, the engine will turn on :-
 (A) 10 minute (B) 5 minute (C) 1.125 minute (D) 2.25 minute
8. Suppose at $t = 0$, the engine turns on, after how much time interval, again, the engine will turn off :-
 (A) 10 minute (B) 5 minute (C) 1.125 minute (D) 2.25 minute

Space for Rough Work

Paragraph for Questions 9 and 10

End A of a rod AB of length $\ell_0 = 5\text{m}$ and cross-section area $S = 1\text{m}^2$ is maintained at some constant temperature. The heat conductivity of the rod is varying with the distance x from, the end A as $K = K_0[1 + ax]$, where $K_0 = 1 \frac{\text{Watt}}{\text{mK}}$ and $a = 0.2 \text{ ms}^{-1}$. The other end B of this rod is radiating energy at the rate of $P = 4560 \text{ J/s}$ and the wavelength with maximum energy density emitted from this end is $14500 \text{ \AA}$. (Assume that except the ends, the rod is thermally insulated. Wein's constant $b = 2.9 \times 10^{-3} \text{ m-K}$.)

9. The variation of temperature as a function of x can be expressed as :

(A) $T = \frac{P}{K_0 a S} \ln \left(\frac{1 + a\ell_0}{1 + ax} \right) + T_B$

(B) $T = \frac{P}{K_0 a S} \ln \left(\frac{2 + a\ell_0}{2 + ax} \right) + T_B$

(C) $T = \frac{P}{K_0 a S} \ln \left(\frac{1 - a\ell_0}{1 - ax} \right) + T_B$

(D) $T = \frac{P}{K_0 a S} \ln \left(\frac{a\ell_0}{x} \right) + T_B$

10. The temperature of the end A is equal to :- (Use : $\ln 2 = 0.7$)

(A) 10000 K

(B) 17960 K

(C) 25000 K

(D) None of these

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

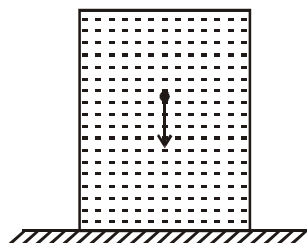
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

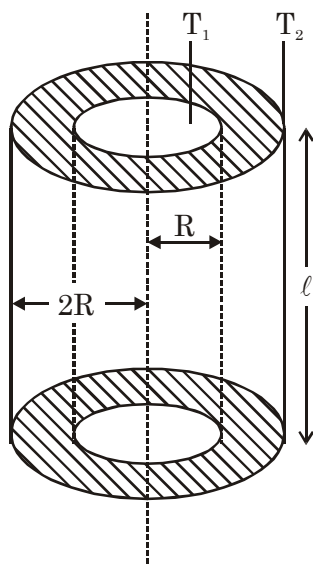
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A dense liquid is completely filled in a closed large container and kept at rest on a horizontal surface. A steel ball is moving downward with a constant speed of 0.5 cm/sec. Find the momentum of dense liquid in CGS unit, if density of liquid is 3 gm/cm³. Given that volume of steel ball is 1.5 cm³.



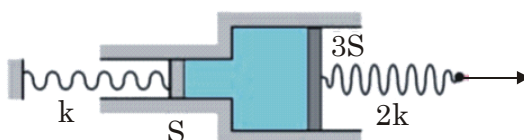
Space for Rough Work

2. Inner surface of a cylindrical shell of length ℓ and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that $(T_1 > T_2)$ as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and $2R$ respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\ell/2$, inner and outer radii $R/4$ and R respectively and thermal conductivity of material nk . If rate of radial outward heat flow remains same for same temperature of inner and outer surface i.e. T_1 and T_2 , then find the value of n .

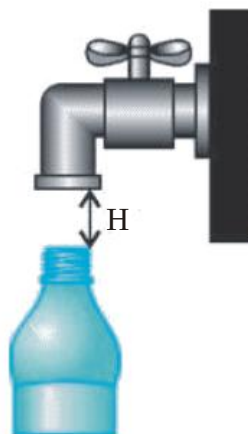


Space for Rough Work

3. In the figure shown, the springs of spring constant k and $2k$ are both at natural length. There is an incompressible fluid between the pistons. Left end of the spring of stiffness k is held fixed. To the right of free end of spring of constant $2k$ we apply an external force and slowly shift this end of the spring by a distance L . How much (in cm) does the other spring stretch? Assume that the fluid under the pistons is not leaking, no friction, and external pressure is atmospheric. The effect of gravity is neglected. Take $p_0 = 10^5$ Pa, $S = 10$ cm², $k = 100$ N/m, $L = 22$ cm.

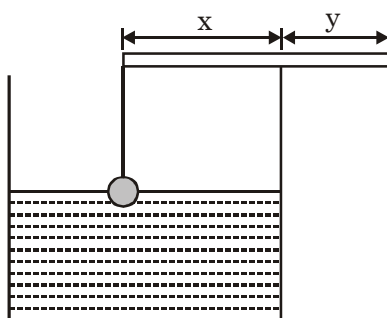


4. We have a bottle with neck of inner diameter of $D = 1$ cm located at a distance of $H = 75$ cm below the water tap. The inner diameter of the tap is $D_0 = 2$ cm (Fig.). The bottle and tap have same vertical axis of symmetry. At what maximum rate of water flow rate Q (in litre/s) will all water get into the bottle? Assume that the water flow is calm (laminar). The effect of air displaced from the bottle in the process of filling with water is neglected.



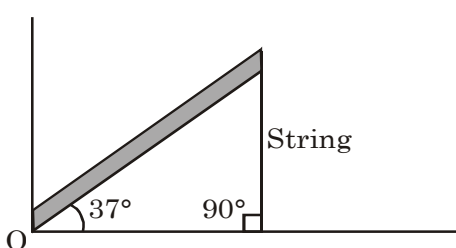
Space for Rough Work

5. A wire is made by attaching two segments together end to end. One segment is made of aluminium and other is steel. The effective linear expansion of two segment is $17 \times 10^{-6}/^{\circ}\text{C}$. What is the ratio of length of aluminium to total length of wire. (Linear coefficients of thermal expansion of aluminum and steel are $23 \times 10^{-6}/^{\circ}\text{C}$ and $11 \times 10^{-6}/^{\circ}\text{C}$ respectively).
6. A homogeneous aluminium ball of radius $r = 0.5 \text{ cm}$ is suspended on a weightless thread from an end of a homogeneous rod of mass $M = 4.4 \text{ g}$. The rod is placed on the edge of a container with water so that half the ball is submerged in water when the system is in equilibrium as shown. The densities ρ_{al} and ρ_{w} of aluminium and water are $2.7 \times 10^3 \text{ kg/m}^3$ and $1 \times 10^3 \text{ kg/m}^3$ respectively. Determine the ratio y/x of the parts of the rod to the brim, neglecting the surface tension on the boundaries between the ball and water. (Use : $\pi = 3.14$)



Space for Rough Work

7. A non-viscous liquid of constant density 10^3 kg/m^3 flows in streamline motion along a vertical tube PQ of variable cross-section. Height of P and Q are 2m and 2.5m respectively. Area of tube at Q = 3 times area of tube at P. Find the work done per unit volume (in KJ/m^3) by pressure as liquid flows from P to Q. Speed of liquid at P is 3 m/s ($g = 10 \text{ m/s}^2$).
8. A cylindrical rod of uniform cross-section, is attached at O in a water tank. The linear mass density of rod is $\lambda_0 x$, where x is distance of the element of the rod, from end O as shown in figure. If the tension in string is given by $\frac{10^4}{P} \text{ N}$ then P is (Length of rod 1m, radius of area of cross section is $\frac{1}{\sqrt{\pi}} \text{ m}$, $\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ ms}^{-2}$, $\lambda_0 = 10^3$ in SI unit). Rod is completely submerged in water.



Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1 For the reaction



what is the % of NH_3 converted if the mixture diffuses twice as fast as that of SO_2 under similar conditions.

- (A) 3.125 % (B) 31.25 % (C) 6.25 % (D) 62.5 %

2. Read the below characteristics carefully.

- (i) Atleast one d-orbital is participating in hybridisation of central atom
- (ii) Atleast four atoms are lying in one plane
- (iii) At least three atoms are in one line
- (iv) At least two planes are present having maximum number of atoms in one plane of respective molecule
- (v) Atleast one lone pair on the central atom is present

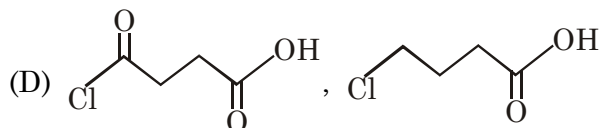
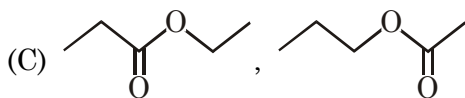
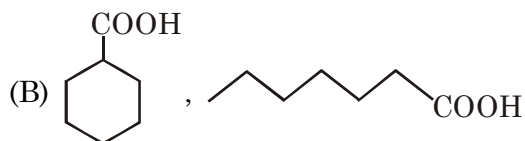
Choose the species for which atleast three characteristics will match :

- (A) CO_2 (B) SO_3F^- (C) $[\text{XeO}_6]^{4-}$ (D) XeO_3F_2

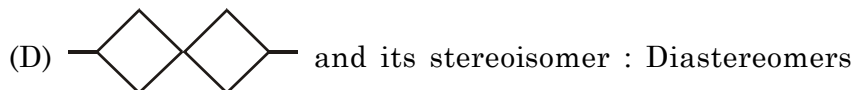
Space for Rough Work

3. Which of the following pairs of the compounds don't have the same root word in their systematic IUPAC names?

(A) Isooctane, n-pentane



4. Ether is considered a good solvent for solvent extraction of organic compounds due to :
 (A) Less polarity of ether (B) Less boiling point of ether
 (C) Less reactivity of ether (D) Higher solubility of organic compounds in ether
5. Which of the following is/are the incorrect relation(s) between the given pair of compounds?
 (A) Methane nitrile and methaneisocyanide : functional isomers
 (B) N-ethylpentanamine and N-ethyl-N-methylbutanamine : Chain isomers
 (C) Mesotartaric acid and (2R, 3R)-tartaric acid : Epimers



6. Which of the following is/are the correct order(s) for the stability of conformers of the following compounds?
 (A) n-butane : Anti > Gauche
 (B) 2-Fluoroethanol : Gauche > Anti
 (C) Ethane-1,2-diol : Gauche > Anti
 (D) Cyclohexane : Chair > Boat > Twisted boat > Halfchair

Space for Rough Work

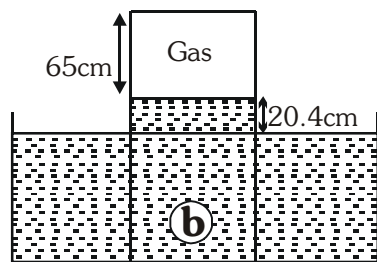
SECTION-I(ii) : (Maximum Marks : 16)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 7 and 8

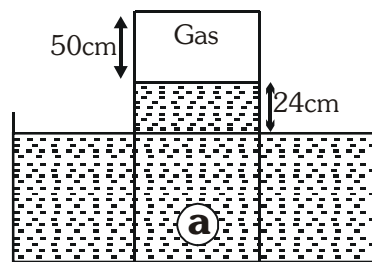
If same amount of gas is trapped over liquid @ and liquid ⑥ in following containers. Assuming temperature and cross-sectional area of container are same.

$$P_{\text{atm}} = 760 \text{ mm Hg} ; d_{\text{Hg}} = 13.6 \text{ gm/mL}$$



V.P. of liquid ⑥ = 110 mm Hg

Density = —



V.P. of liquid of ⑥ = 50 mm Hg

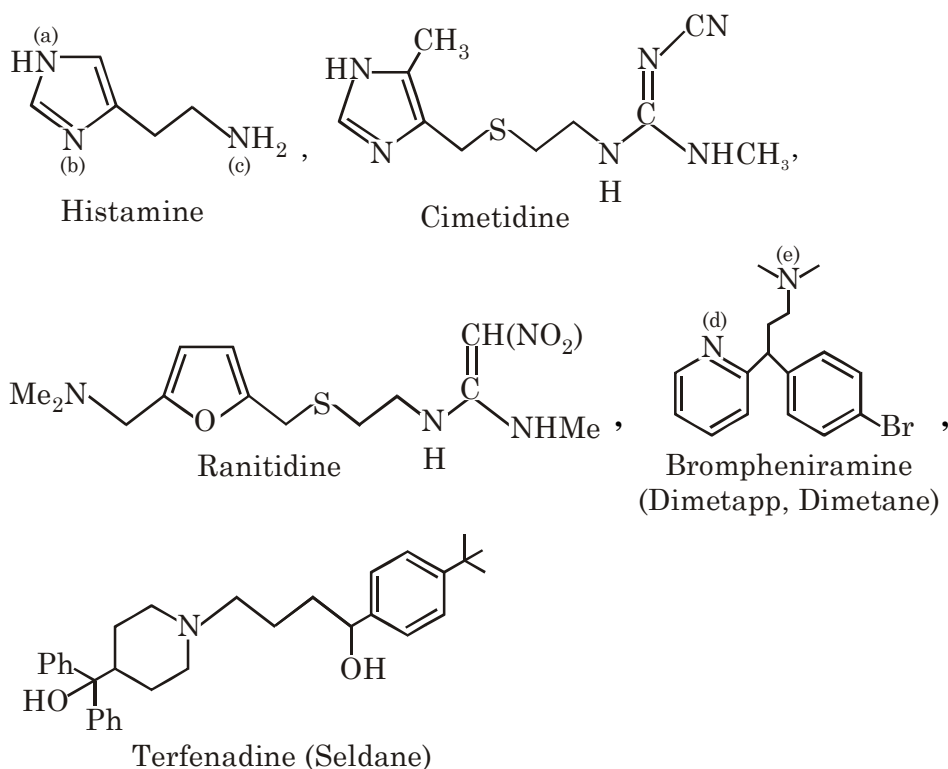
Density = 3.4 g/mL

- Find the pressure (in cm Hg) of gas over liquid ⑥.
 (A) 76 (B) 11 (C) 50 (D) 65
- Find the pressure (in cm Hg) of gas over the liquid @.
 (A) 76 (B) 11 (C) 50 (D) 65

Space for Rough Work

Paragraph for Questions 9 and 10

A major breakthrough in the treatment of hyperacidity came through the discovery according to which a chemical, histamine, stimulates the secretion of pepsin and hydrochloric acid in the stomach. The drug cimetidine (Tegamet), was designed to prevent the interaction of histamine with the receptors present in the stomach wall. This resulted in release of lesser amount of acid. The importance of the drug was so much that it remained the largest selling drug in the world until another drug, ranitidine (Zantac), was discovered. Synthetic drugs, brompheniramine (Dimetapp) and terfenadine (Seldane), also act as antihistamines.



9. Which of the following statements are incorrect about the given compounds?
- The total number of chiral carbon atoms in histamine, cimetidine and ranitidine combined is 0.
 - The DOU of tegamet is 4.
 - There are no tertiary carbon atoms in ranitidine.
 - Histamine, cimetidine and ranitidine are all examples of aromatic compounds.
10. Which of the following statements is correct regarding given compounds?
- The DOU of terfenadine is 5
 - The compound brompheniramine is optically inactive.
 - Order of basic strength of N in histamine : b > a
 - One mole of terfenadine reacts with NaOH to form a disodium salt

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

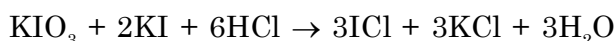
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	●
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A definite volume of pure ammonia (NH₃) gas is passed through a series of electric sparks by which the volume becomes 90 mL. The increase in volume is due to formation of nitrogen (N₂) and hydrogen (H₂) gases. All the gases finally present are washed with dilute sulphuric acid solution by which volume of gases becomes 80 mL. All the volume are measured at the same pressure and temperature. The percentage of NH₃ gas decomposed into N₂ and H₂ gases is ?
2. 1 gm sample of AgNO₃ is dissolved in 50 mL of KI solution. The AgI precipitated is filtered off. Excess of KI in filtrate is titrated with M/10 KIO₃ in presence of 6M HCl till all I⁻ converted into ICl. It requires 50 mL of M/10 KIO₃ solution. 20 mL of same stock solution of KI requires 30 mL of M/10 KIO₃ under similar conditions. Calculate % of AgNO₃ in sample reaction is.



Space for Rough Work

3. Total number of species which have either bond length or bond angle higher than atleast one other species given –

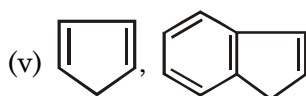
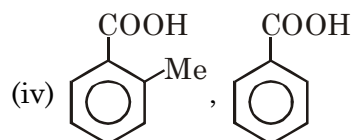
BF_3 , BCl_3 , NH_3 , H_2O , H_2S , XeF_2

4. Select total number of pairs out of following in which first compounds isomere acidic than second

(i) Picric acid, Phenol

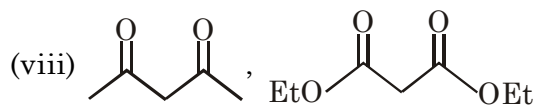
(ii) Acetylene, Toluene

(iii) HI , HBr



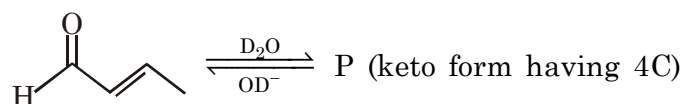
(vi) H_2S , H_2O

(vii) Ethanoic acid, Ethanamide



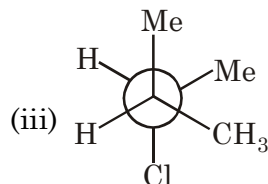
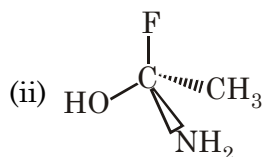
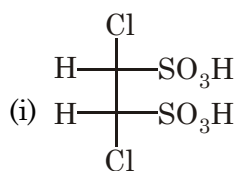
Space for Rough Work

5. Consider the following reaction



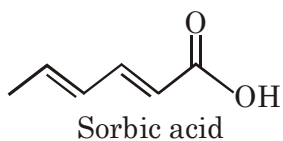
Number of deuterium atoms in 1 molecule of P after long time is?

6. Total number of carbon atoms with "R" configuration in the following compounds is?



Space for Rough Work

7. Salts of sorbic acid and propanoic acid are used as food preservatives (prevent microbial growth on food). What is the total number of stereoisomers for sorbic acid?



8. Total monochloro derivatives possible for 1,4-dichlorocyclohexane is (X), fractions obtained on fractional distillation of all monochloro derivatives is (Y) and total enantiomeric pairs in all monochloro derivatives is (Z). Find value of $(X) \times (Y) \times (Z)$.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Consider parabola $y^2 = 4x$. Then locus of centroid of a triangle formed by vertex of parabola and a variable focal chord is a conic
 (A) whose eccentricity is $\sqrt{2}$ (B) whose eccentricity is 1
 (C) whose length of latus rectum is $\frac{3}{4}$ (D) whose length of latus rectum is $\frac{4}{3}$
2. Given the inequality $\frac{1}{3} \leq \frac{x^2 - \alpha x + \alpha^2}{x^2 + \beta x + \beta^2} \leq 3$ holds true $\forall x \in \mathbb{R}$ and $\beta \neq 0$, then which of the following can be true ?
 (A) $\frac{\alpha}{\beta} \in [1, 3)$ (B) $\frac{\alpha}{\beta} \in \left(-2, -\frac{1}{2}\right)$ (C) $\frac{\alpha}{\beta} \in (3, 5)$ (D) $\frac{\alpha}{\beta} \in (-5, -2)$

Space for Rough Work

3. If the straight line $y = x$ intersects the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ at 2 distinct points A and B, such that angle between tangent at A and the line is θ , then
- (A) $\cos \theta = \frac{1}{4\sqrt{2}}$ (B) $\tan \theta = \frac{2}{\sqrt{3}}$
- (C) length of the chord $AB = \sqrt{62}$ (D) length of the chord $AB = \sqrt{59}$
4. Let a variable circle touches a line, which is tangent to the parabola $y^2 = 4x$ at $(4, 4)$ and variable circle passes through focus of the parabola $y^2 = 4x$. If locus of its centre is conic 'C', then
- (A) directrix of conic 'C' is $x - 2y + 4 = 0$ (B) focus of conic 'C' is $(4, 0)$
- (C) latus rectum of conic 'C' is $2\sqrt{5}$ (D) axis of the conic 'C' is $2x + y - 2 = 0$.
5. If a straight line passes through the point $P(2, 0)$ having equation $y = m(x - 2)$ intersects the curve $y^2 = 8x$ at the points Q, R (let $PQ = \ell_1$ & $PR = \ell_2$). Also given that G, A represents geometric and arithmetic means of ℓ_1 and ℓ_2 respectively, then correct option(s) is/are
- (A) $\ell_1, \ell_2 \geq 2$ (B) $\ell_1, \ell_2 < 2$ (C) $G \propto A$ (D) $G \propto \sqrt{A}$
6. If in an acute angle $\triangle ABC$, the line joining the circum centre O and incentre I is parallel to BC, then
- (A) $\cos B + \cos C = \frac{1}{2}$ (B) $\cos B + \cos C = 1$
- (C) $\frac{\cos A}{\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}} = 4$ (D) $\cos A = \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 16)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 7 and 8

Given pair of lines $2x^2 + 5xy + 2y^2 + 4x + 5y + a = 0$ and the line L as $bx + y + 5 = 0$.

On the basis of above information, answer the following questions :

7. If there exist four circle which touch pair of lines and the line L simultaneously, then the value of b can be -
(A) 3 (B) 5 (C) 7 (D) 9
8. If there exist 2 circles which touch pair of lines and line L simultaneously, then the possible value of b can be -
(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 2 (D) $\frac{5}{2}$

Space for Rough Work

Paragraph for Questions 9 and 10

If $a_1, a_2, a_3 \dots a_n$ are positive real numbers and A, G and H denotes arithmetic mean, geometric mean and harmonic mean of $a_1, a_2, \dots, a_n$ respectively, then $A \geq G \geq H$

and equality exist if $a_1 = a_2 = a_3 = \dots = a_n$.

On the basis of above information, answer the following questions :

9. If a_1, a_2, a_3 are the lengths of sides of a scalene triangle, then value of $(a_1 + a_2 - a_3)(a_2 + a_3 - a_1)(a_3 + a_1 - a_2) - a_1 a_2 a_3$ can be
 (A) 1 (B) 2 (C) -1 (D) -2
10. If $a_1, a_2, a_3, \dots, a_n$ are in arithmetic progression with common difference $d \neq 0$ also $S = \sum_{r=1}^n a_r$,
 then which of the following can be true ?

(A) $S > \frac{n^2}{\left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}\right)}$

(B) $S > \frac{2n.a_1 a_n}{a_1 + a_n}$

(C) $S > n\sqrt{2a_1(n-1)d}$

(D) $S < n\sqrt{a_1 a_n}$

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	●	●	●	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	●	●	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. $y + x = 0$ and $y - x = 0$ are tangents at the vertex and axis of a parabola respectively and $y = 2x + 3$ is a tangent to the same parabola. Then the square of length of latus rectum is
2. If $x, y, z \in (0, 1)$ such that $x + y + z = 1$, then least value of $\frac{(1-x)(1-y)(1-z)}{xyz}$ is

Space for Rough Work

3. Let $\begin{vmatrix} x & 2 & x \\ x^2 & x & 6 \\ x & x & 6 \end{vmatrix} = Ax^4 + Bx^3 + Cx^2 + Dx + E$, then the value of $(5A + 4B + 3C + 2D + E)$ is equal to

4. $\frac{\cos 6^\circ}{(\cos 14^\circ + 2\cos^2 7^\circ)\cos 83^\circ \cos 21^\circ \cos 42^\circ}$ is equal to

Space for Rough Work

5. A regular hexagon is drawn circumscribing the circle with centre $\left(1, \frac{1}{\sqrt{3}}\right)$ and radius 2. $(3, \sqrt{3})$ is one vertex and other vertices are $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$ and (x_5, y_5) . Then the value of $\sum_{i=1}^5 (x_i + y_i)$, is
6. Let P and Q are two distinct points on circle $x^2 + y^2 + 2gx + 2fy + c = 0$ such that circle have minimum radius. Consider P and Q are two points from which we can draw perpendicular tangents on both circles $x^2 + y^2 - 32 = 0$ & $x^2 + y^2 - 4x - 4y - 10 = 0$ then the value of $g + f + c$ is

Space for Rough Work

7. If $(x^2 + 1)(y^2 + 1) + 9 = 6(x + y)$ (where $x, y \in \mathbb{R}$), then the value of $(x^2 + y^2)$ is
8. Let system of equations in two variables x, y such that $x^2 + y^2 - 2x + 6y - 15 = 0$ and $x^2 + y^2 - 12x - 18y + c = 0$ have two common solutions, then number of positive integral value(s) of c is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	24
I(ii)	Paragraph Based (One or more correct option(s))	4	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	16
II	Numerical Value Type (Up to second decimal place)	8	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	32

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator